

Tests specific to aerospace and defence

Certain technologies such as structural testing, radar and cryogenics are specific to this segment. The aerospace & defence industry itself creates test equipment and simulation environments or works with T&M manufacturers to create specific solutions. The aircrafts and space vehicles have to undergo structural tests to ensure correct behaviour under extreme conditions. Communications and security and surveillance are important part of the aerospace and defence industry, and their requirements are very specific.

Structural tests. An aircraft undergoes several tests before it is considered ready for flight. Structural tests are performed on the airframe and the requirements are really complex. There needs to be tight timing and synchronisation, high channel counts, real-time and post-acquisition signal processing, advanced control algorithms, data management and advanced visualisation. To fulfil these requirements, you need an extremely advanced test system that offers extensive engineering resources, and can be expanded and



Agilent handheld RF vector network analyser

customised to meet the changing requirements. At the same time, it needs to be cost-effective.

Semancik shares, "Equipment available today have set a new standard of performance for stress and fatigue testing. Independent high-resolution analogue-to-digital converters, extensive software-selectable filtering and independent signal conditioning paths provide exceptional accuracy

and reliability." Many equipment deliver cost-effective price per channel performance with both high-quality static and high-speed strain gauge measurements in critical test applications.

"When it comes to satellite and its sub-systems, "getting it right the first time" takes on a larger meaning with satellites going into earth orbit or travelling into deep space," says Awasthi. A combination of software-simulated environment and hardware is used to ensure the quality and reliability of the device under test.

RF and microwave tests. The primary requirement for radio frequency and microwave test, apart from accuracy and precision, is wide-band coverage. T&M manufacturers offer highly flexible software along with modular RF hardware to support a wide range of RF test applications that include signal intelligence, record and playback, software-defined radio, multiple-input multiple-output and beam-forming.

Test equipment use some of the latest and high-performance technologies like field-programmable gate arrays for signals intelligence applications. For communications and R&D-grade accuracy, RF equipment are available that provide multichannel synchronisation for up to 4x4 MIMO configurations and a wide range of communication IPs for software-defined radio applications.

High-performance signal analysers provide frequency coverage up to 26.5 GHz and ensure present and future flexibility through optional measurement capabilities. Hardware expandability is another key feature of such equipment. High-speed digitiser modules provide measurement fidelity for high-frequency signals. These also feature large-scale circuit integration, high-speed GS/s sampling rates, wide bandwidth and large acquisition memories.

Surveillance tests. "The growth in types of long-range and short-range wireless technologies poses an escalating challenge for intelligence and homeland

India's aerospace & defence industry

The Stockholm International Peace Research Institute (SIPRI) Yearbook 2010 ranks India ninth in the list of the world's top 15 military spenders in 2009. India is also one of the six countries in the world that undertake space launch. Certainly, India's aerospace and defence industry is growing rapidly, ever since the government initiated its open skies policies.

"In 2001, the government allowed 100 per cent domestic private investment in the defence sector upon obtaining an industrial licence and FDI of up to 26 per cent. Allowing private investment has begun to attract a number of Indian companies into the sector. The introduction of the defence offset policy in 2006 and significant liberalisation in 2008 has provided significant opportunities for Indian companies entering the sector," shares Gautam Awasthi, general manager, marketing, Agilent Technologies.

"As the culture of indigenous designs grows in India, T&M manufacturers are excited and looking forward to provide latest solutions for safe and high-quality developments," says Jon Semancik, director of marketing, VTI Instruments.

"India is consistently investing in defence and most of the companies look at India as a huge market. Being a developing country, it is also a critical market. Innovation with cost-effectiveness can be a better solution in India," says Nareish Narasimhan, Pacific technical marketing manager, Tektronix.

"Leading companies like Boeing and Airbus have their own maintenance, repair and overhauling facility in India. But we are still in prototype stage as far as our own aircraft design is concerned. It will be a great achievement when Thejas/LCA aircrafts fly on our airspace," shares S.R. Sabapathi, chief executive officer, QMAX Technologies.